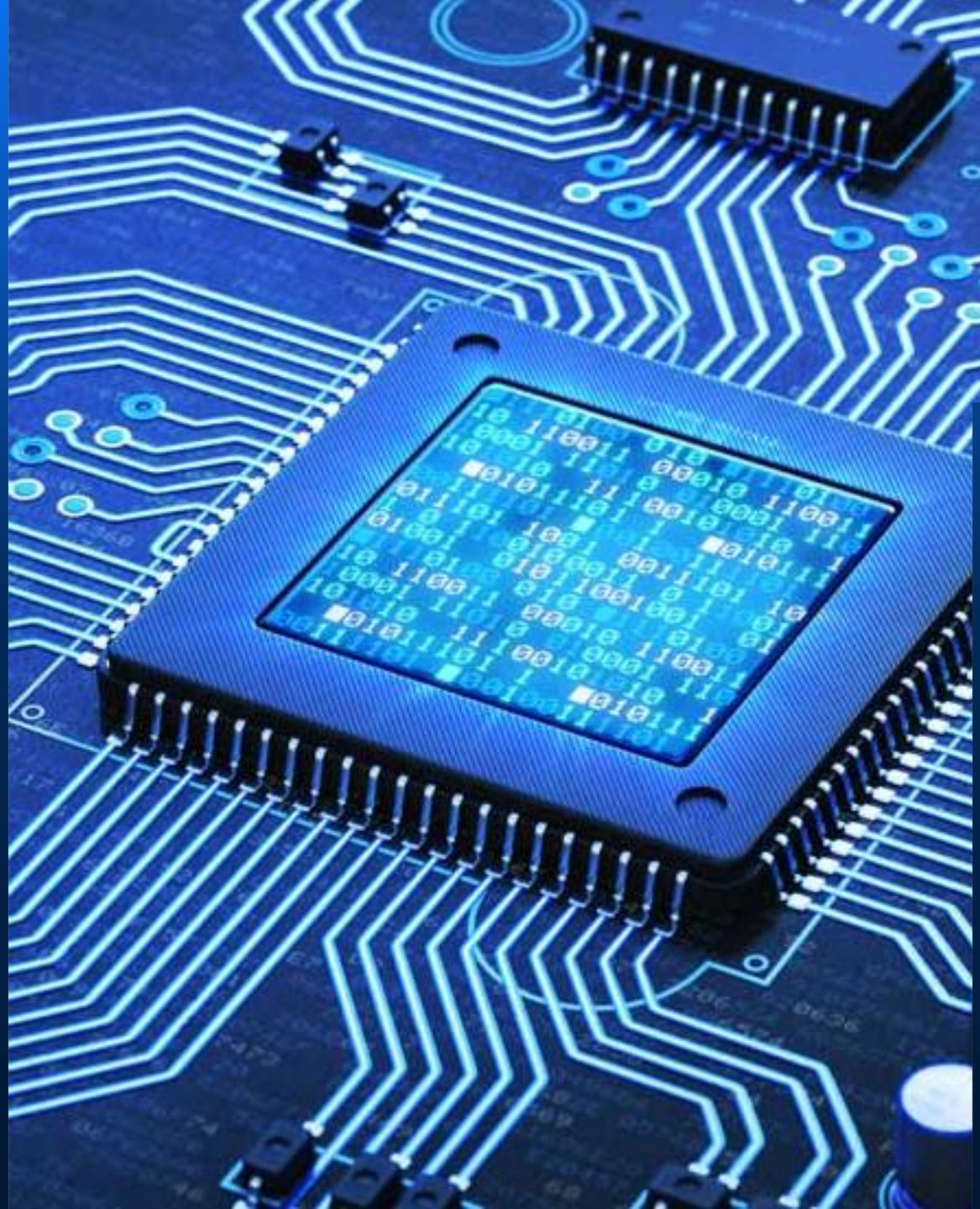


# Computer Architecture

## Practice Session 6

By: Eng. Abdulkareem Mohammed

Senior HW Digital Design Engineer

# Agenda

1. Memory using BRAM

# 1. Memory using BRAM

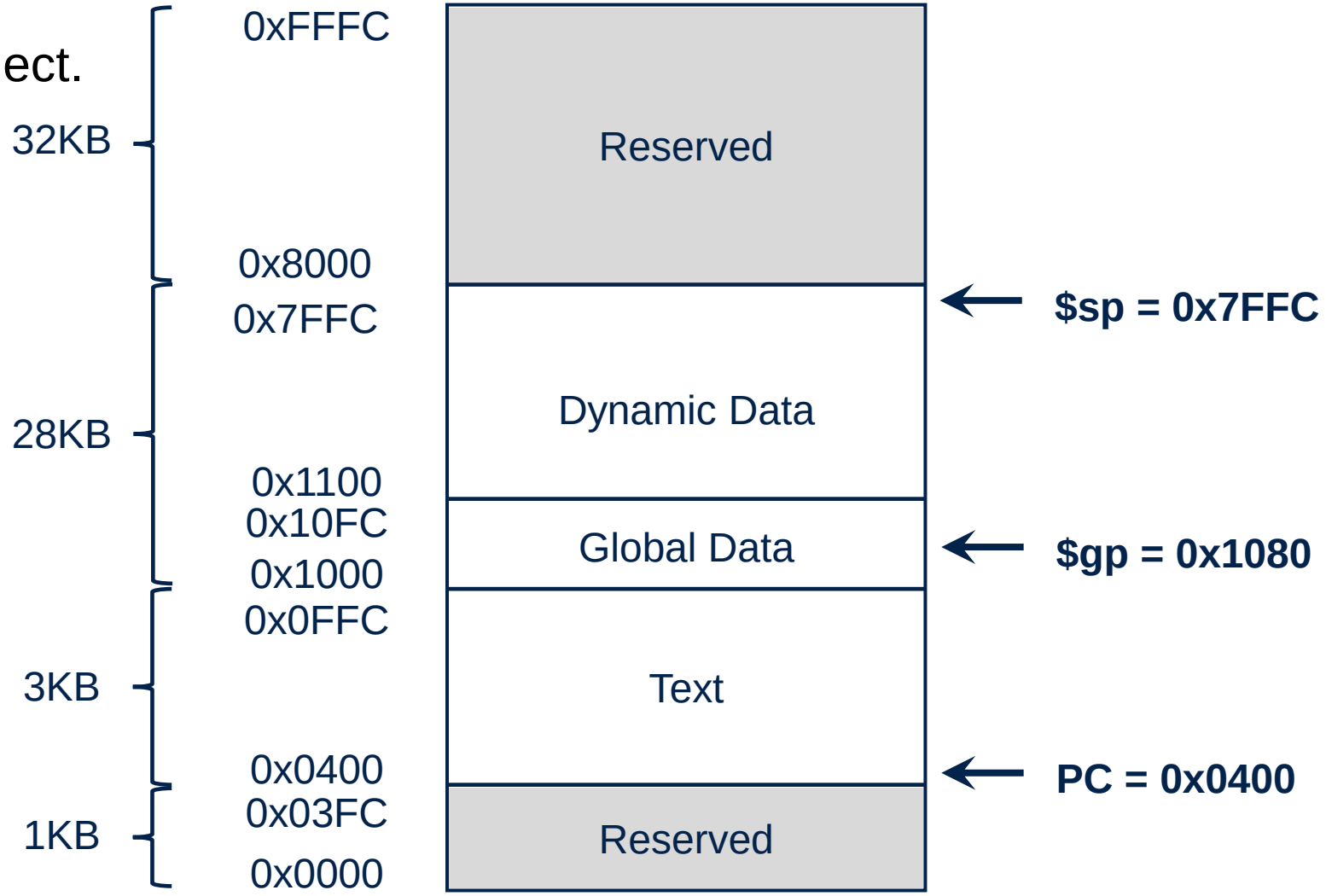
# MIPS ports

- Define the ports of MIPS

```
module mips (
        // Global
        input           CLK ,
        input           RST ,
        // Instruction memory port
        input [3:0]     data_in,
        input           wr_en,
        //output
        output  [3:0]   s0_reg,
        //interrupt
        input           itr
    );
```

# Memory using BRAM

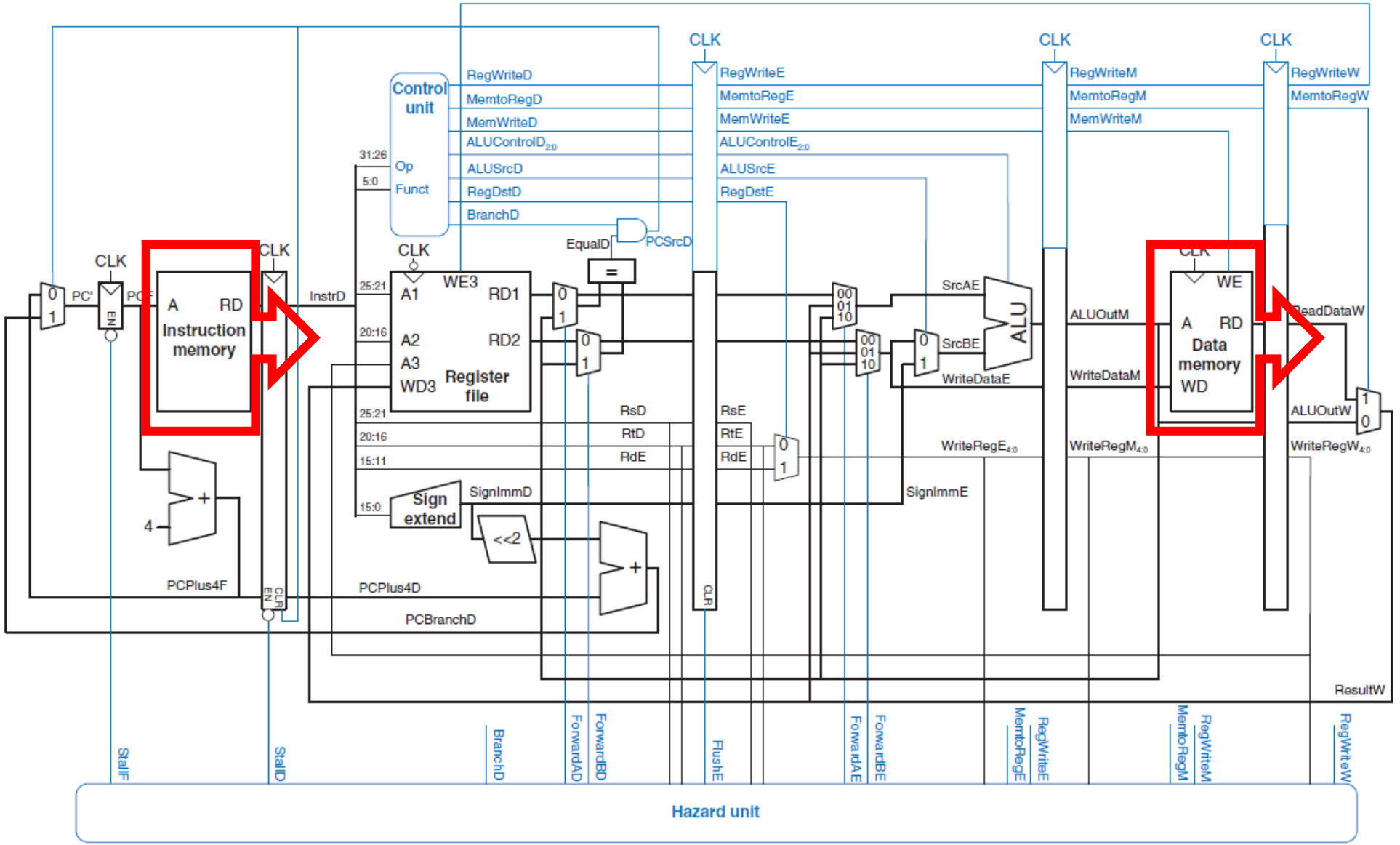
- Memory Model used for Our Project.




**Exception Handler Address = 0x8180**

# Memory using BRAM

- .



# Memory using BRAM

- .

```
parameter SIZE = 28*1024/4;     // 28 KB
 // --- Memory Array ---
 reg [7:0] BRAM0 [0:SIZE-1];
 reg [7:0] BRAM1 [0:SIZE-1];
 reg [7:0] BRAM2 [0:SIZE-1];
 reg [7:0] BRAM3 [0:SIZE-1];

// --- Infer RAM ---
 always @ (posedge HCLK)
 begin
   if (wr_en[0])      BRAM0[addr] <= wr_data[7:0];
   if (wr_en[1])      BRAM1[addr] <= wr_data [15:8];
   if (wr_en[2])      BRAM2[addr] <= wr_data [23:16];
   if (wr_en[3])      BRAM3[addr] <= wr_data [31:24];

   // do not use enable on read interface.
   rd_data <= {BRAM3[addr],BRAM2[addr],BRAM1[addr],BRAM0[addr]};
 end
```